9 Fluorine-18 ($^{18}_{9}F$) is a radioactive nuclide that is used as a tracer in positron emission tomography (PET scanning). Fluorine-18 decays to a nuclide of oxygen (O) according to

$$^{18}_{9}F \longrightarrow {}^{Q}_{P}X + {}^{R}_{8}O.$$

(a) (i) State what is meant by a tracer.

.....

..... [1]

(ii) State the symbol of the particle that is represented by X and the values of P, Q and R.

X: ..... P: .....

Q: ..... R: .....

[2]

(b) (i) Explain how the radioactive decay of fluorine-18 results in the emission from the body of the gamma-ray photons that are detected during a PET scan.

.....

.....

.....

..... [2]

(ii) Explain how the detection of the gamma-ray photons is used to produce an image of the tissue being examined.

.....

.....

.....

..... [2]

(c) The half-life of fluorine-18 is T.
A patient is injected with amount of substance n of fluorine-18.

(i) Determine an expression for the initial value R_0 of the rate R of production of gamma-ray photons by the tracer, in terms of n, T and the Avogadro constant N_A.

R_0 = ... [3]

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(ii) On Fig. 9.1, sketch the variation with time t of R.

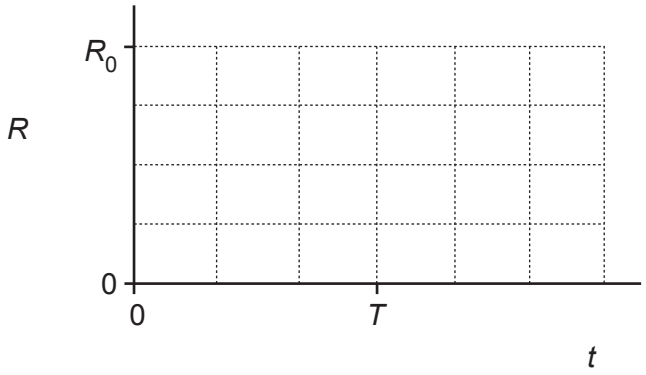


Fig. 9.1

[2]

[Total: 12]